

Week 4 Make controls reliable · paper route

Paper route · GROW Computing · 40 minutes

Name: _____ Date: _____

Week 4 Paper Route. Make controls reliable.

I can create and test all four arrow controls in my maze.

You can finish the whole of Week 4 on this paper.

This is the whole week. It is not a practice.

Paper and screen are two roads to the same award.

You do not need a computer today.

You need this sheet, a pen and a counter.

A counter can be a coin, a rubber, or a folded square of paper.

Take out your Week 3 paper maze. Put it beside this sheet.

If you do not have it, use the maze printed here.

Point, say, draw or write. All of these count.

An adult can write your exact words for you.

Today you will write four key scripts.

Then you will move your counter to test each one.

Your teacher watches you do it and writes down what you did.

Nothing you do today has to be done again on a computer.

Point, say, draw or write. Someone may record your exact words.

On the screen route this week is saved as W04_Controls.sb3. On the paper route no file is made. Your pages are the record: write your name and the date on every page and put them in your folder. An adult initials the folder. Nobody should ask you for W04_Controls.sb3 if you took the paper route.

Write your code

Player · Reset. This script puts Player back on your safe start when the game begins. Write the x and y of your own start square in the block. Point to the hat block. Say what starts this script.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- go to x (___) y (___)
- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Right control. Right is across. Across is x. Choose one block from the word bank and write it in row 1. Point to the hat block. Say the key name out loud before you write.

Write one block on each line. Start with the hat block, which is done for you.

when right arrow key pressed

Blocks you may use this week:

- go to x (___) y (___)
- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Left control. Left is across. Across is x. Left goes back, so the number is below zero. Choose one block from the word bank and write it in row 1.

Write one block on each line. Start with the hat block, which is done for you.

when left arrow key pressed

Blocks you may use this week:

- go to x (___) y (___)
- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Up control. Up is not across. Up is y. This is the script with the fault on the fault card. Choose the block that changes y, not x.

Write one block on each line. Start with the hat block, which is done for you.

when up arrow key pressed

Blocks you may use this week:

- go to x (___) y (___)
- change x by 10
- change x by -10
- change y by 10
- change y by -10

Player · Down control. Down is y. Down goes back, so the number is below zero. Choose one block from the word bank and write it in row 1.

Write one block on each line. Start with the hat block, which is done for you.

when down arrow key pressed

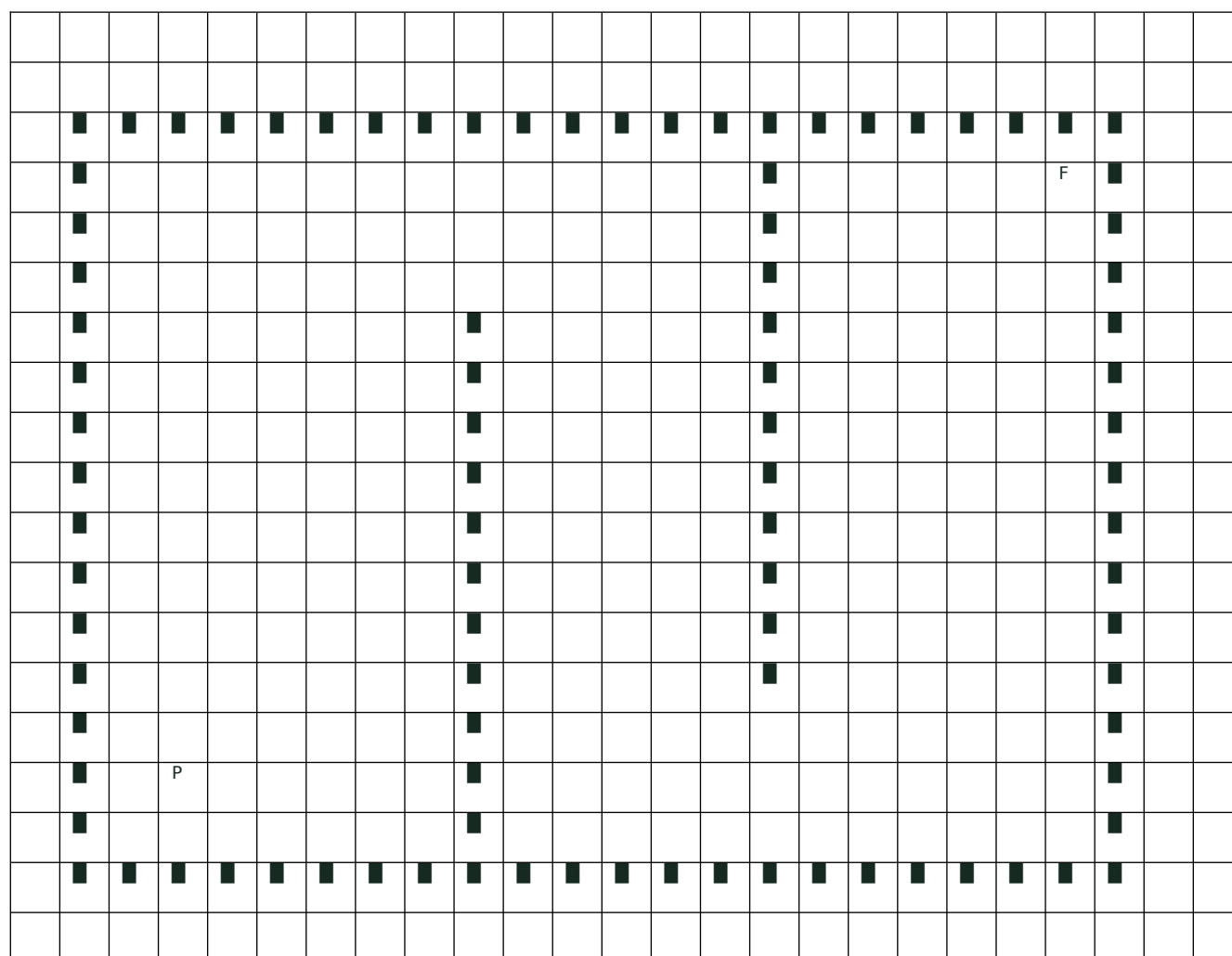
Blocks you may use this week:

- go to x (___) y (___)
- change x by 10
- change x by -10
- change y by 10
- change y by -10

The maze

The maze is printed at its real size. The stage is 480 across and 360 up. x runs from -240 on the left to 240 on the right. y runs from -180 at the bottom to 180 at the top. Every square on the grid is 10 across and 10 up. One key press moves the counter one square. The numbers under the grid give you x. The numbers beside the grid give you y. Six walls are shaded grey. Wall 1 is the floor: x -220 to 220, y -160 to -140. Wall 2 is the roof: x -220 to 220, y 140 to 160. Wall 3 is the left side: x -220 to -200, y -140 to 140. Wall 4 is the right side: x 200 to 220, y -140 to 140. Wall 5 stands up from the floor: x -70 to -50, y -140 to 80. Wall 6 hangs down from the roof: x 50 to 70, y -80 to 140. Here is what you do on the grid. 1. Put your counter on a square that is not grey. 2. Read its x from the numbers under the grid. 3. Read its y from the numbers beside the grid. 4. Write that x and that y in your go to block. 5. Write the same x and y in the My safe start box. That square is now your start. 6. Reset means put the counter back on your start square. 7. Reset. A partner or an adult says one key name. Move the counter one square for that key. 8. Write the new x and the new y in the trace table. 9. Reset. Do it again for the next key. Do all four keys. 10. Reset one more time. Now do right, up, left, down without stopping. 11. The counter must land back on your start square. 12. Walls do not stop the counter this week. Going through a wall is fine in Week 4. Walls stop Player in Week 5.

The maze below is the real one, at its real size. One square is 20 steps. Player starts at (-180, -120). Finish is at (180, 120). Move a counter square by square and write down where it lands.



Predict, trace, compare

Predict first. One key press from the start. Do not move the counter yet.

I press this key	The block in my script	Which way will Player move?	New x	New y
green flag	go to x -180 y -120	Player goes to the start	-180	-120
right arrow				
left arrow				
up arrow				
down arrow				

Trace 1. Model start x -180, y -120. Reset before every press. Write the x and y after each block.

Step	Block that runs	x after	y after
1	green flag: go to x -180 y -120	-180	-120
2	right arrow: change x by 10		
3	green flag: go to x -180 y -120	-180	-120
4	left arrow: change x by -10		
5	green flag: go to x -180 y -120	-180	-120
6	up arrow: change y by 10		
7	green flag: go to x -180 y -120	-180	-120
8	down arrow: change y by -10		

Trace 2. The return. Reset once, then right, up, left, down with no reset in between.

Step	Block that runs	x after	y after	Same square as my start?
1	green flag: go to x -180 y -120	-180	-120	yes

2	right arrow: change x by 10			
3	up arrow: change y by 10			
4	left arrow: change x by - 10			
5	down arrow: change y by - 10			

Trace 3. My own start. Use the x and y you wrote in your own go to block.

Step	Block that runs	x after	y after
1	green flag: go to x ___ y ___		
2	right arrow: change x by 10		
3	up arrow: change y by 10		
4	left arrow: change x by -10		
5	down arrow: change y by -10		
Check	Step 5 must be the same as step 1. Tick this box if it is.		

Fault card. Compare two scripts for the up key. Trace both. Circle the right one.

Script for the up key	Start x	Start y	x after one press	y after one press	Which way did Player move?
when up arrow key pressed: change x by 10	-180	-120			
when up arrow key pressed: change y by 10	-180	-120			

Circle the script that is right for the up key. Say what was wrong with the other one.					
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Which hat block starts it? Write or say the hat block for each job.

What I want to happen	The hat block I need
Player goes to my safe start when the game begins	
Player moves right	
Player moves left	
Player moves up	
Player moves down	

Challenge. Does the way Player faces change x and y? Draw the arrow on your counter first.

Player is facing	Block I run	Start x, y	After one press x, y	Which way did Player move?
direction 90, arrow points right	change x by 10	-180, -120		
direction 0, arrow points up	change x by 10	-180, -120		
direction 0, arrow points up	move 10 steps	-180, -120		
I kept change x and change y in my four scripts. Tick to show you kept them.				

My completion record

What this week shows

AQA 71638 outcome 2: Observe pupil creation of all four arrow scripts and a practical demonstration. Check any previously inherited controls by having the pupil construct the relevant script.

- ☐ I made four separate key scripts for Player. I wrote each one myself.
- ☐ My right and left scripts change x. My up and down scripts change y.
- ☐ My reset uses a start square that is clear of every wall. I wrote its x and y.
- ☐ I moved my counter for every test. I wrote the new x and y each time.
- ☐ I reset, then did right, up, left, down. My counter came back to my start.
- ☐ I found the fault on the up key card. I said the axis was wrong, not the number.
- ☐ I put my sheets in my folder. I took them out again. I tested my controls again from my own writing.
- ☐ Tick, point to, or say yes for each line. An adult can write your words for you.

Adult record

- ☐ Observed the pupil select Player: the pupil pointed to or named Player from the three sprite headers (Maze, Player, Finish) before writing anything.
- ☐ Observed the pupil place the counter on a square clear of every wall, read its x from the printed x axis and its y from the printed y axis, and write that same pair into the go to x y block. Record the pupil's actual safe start; (-180,-120) is the supplied model example only.
- ☐ Observed the pupil write each of the four key scripts on its own separate grid, choosing the motion block from the word bank. The block was not named for them and was not copied from the teacher answer page.
- ☐ Observed that right and left use change x, and up and down use change y, in the scripts the pupil wrote.
- ☐ Observed the pupil reset the counter to their own start square before each single test, move the counter one square for the key given, and record the resulting x and y. The recorded pairs agree with the script the pupil wrote.
- ☐ Observed the practical demonstration: reset, then right, up, left, down in one sequence, the counter returning to the start square, with the four rows of Trace 2 completed.
- ☐ Observed the pupil repair the printed up-arrow fault by naming the axis (change y), not by changing the sign.
- ☐ Observed the pupil name the hat block for each of the five jobs in the Which hat block starts it? table, so the event was chosen by the pupil and not only read off the printed grid.
- ☐ Observed the pupil take the sheets back out of the folder and repeat one key test from their own writing, unaided.
- ☐ Any inherited or copied work was re-checked by having the pupil construct the relevant script again on a fresh grid, unaided.

- ☐ Support used and communication method (pointed, said, drew, wrote, or exact words scribed) recorded against each row.
- ☐ Route taken: PAPER. Record this here in the teacher guide only. Do not write the route on, or alter, the AQA summary sheet.
- ☐ Record the completed outcome on the AQA summary sheet from this observation, exactly as for a screen route. No later device opportunity is required for this outcome.
- ☐ All four controls remain the target for every route; two keys are partial progress.

Adult name: _____ Date: _____

Route taken: paper · Recorded in the teacher guide, not on the AQA form.